

Lab 3: Architectural Pattern Selection & Component Modelling

Patient Health Record Consent Management System (PHRCMS)

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Architecture Selection: We chose the Layered Architecture for the PHRCMS system.

Reason 1 — Clear separation between access policy and data.

PHRCMS enforces a single, consistent consent policy across three distinct actors (Patient, Clinic Doctor, Clinic Administrator). A Layered Architecture cleanly separates the Presentation Layer (patient/doctor/admin portal), the Business Layer (Consent Management, Doctor Verification, Audit Logging), and the Data Layer (record, consent, and registry storage). Every access decision is forced to pass through the Business Layer, so consent rules (FR-001, FR-002, FR-004) are enforced in exactly one place rather than being duplicated across UI screens.

Reason 2 — Simplicity and traceability for a moderate-scale healthcare workflow.

PHRCMS has a bounded set of well-defined workflows (grant, revoke, view registry, doctor access, doctor verification) rather than independently-scaling, high-traffic services. A Microservices style would add operational complexity (service discovery, distributed transactions) that this scenario does not need, while a Client-Server style would centralize business logic and data together, making the append-only audit requirement (NFR-001) harder to isolate and protect. The layered style keeps the system easy to reason about, test, and map directly back to the functional requirements, while still allowing each layer to be modified independently.

Security Advantage.

Because the Presentation Layer has no direct required interface to the Data Layer, a patient or doctor client can never bypass the Consent Management component to read or write records or audit entries directly. All requests are funneled through Consent Management, which calls Doctor Verification before granting access and calls Audit Logging for every grant, revoke, and access event — satisfying the append-only audit trail requirement (NFR-001) and preventing unverified doctors from being selected for consent (FR-005).

Performance Benefit.

The Consent Management component's required interface to Doctor Verification is a lightweight, in-process status check rather than a network call to a separate service, which keeps the consent-verification path fast. This directly supports NFR-002 (authorize or deny a doctor's access request in under 500 ms), since the layered design avoids the network latency and serialization overhead that a distributed microservices call between these two components would introduce.